A Commuter Route and Fare Information System for Jeepneys and Tricycles in Metro Manila

Christian Gil T. DELOS SANTOS[1], Carlo Jezer L. GARCIA[2],
Juan Miguel Raphael G. GASPAR[3], Malcolm Pyro C. LOPEZ[4]
[1]*Information Systems Program, De La Salle-College of Saint Benilde*
[2]*Information Systems Program, De La Salle-College of Saint Benilde*
[3]*Information Systems Program, De La Salle-College of Saint Benilde*
[4]*Information Systems Program, De La Salle-College of Saint Benilde*

christiangil.delossantos@benilde.edu.ph; carlojezer.garcia@benilde.edu.ph;
juanmiguelraphael.gaspar@benilde.edu.ph; malcolmpyro.lopez@benilde.edu.ph

***Abstract*—**This study addresses the difficulty experienced by the commuters in Metro Manila in identifying jeepney routes, tricycle zones, and expected fares. This research aims to determine the most common commuting problems, assess the need for a route and fare information system, and develop a mobile application solution that improves commute planning and fare transparency. The study gathered data through a survey of commuters. Results showed that many respondents were uncertain about the correct routes to take and the proper fares to pay, especially for tricycles, which often led to confusion and concerns about overpricing. Based on these findings, the study proposes a commuter information system that provides route guidance, fare estimation, jeepney routes, and tricycle zone information in one accessible platform. The system may help commuters make better commuting decisions, reduce fare disputes, and improve transparency in public transportation. Overall, the study highlights the value of a digital tool in supporting safer, clearer, and more efficient commuting.

***Index Terms*—Commuter information system, Fare transparency, Jeepney routes, Mobile App, Tricycle zones**

I. Introduction

Public transportation in urban areas such as Metro Manila, plays a vital role for commuters relying on jeepneys and tricycles [13]. However, unclear routes, tricycle zones, and fares lead to confusion, wasted time, and overcharging risks [7]. This study develops a mobile app to provide route guidance, zone maps, and fare estimates, enhancing commute efficiency [1], [2] [9] [12].

Persistent challenges in Philippine urban transport, particularly with jeepneys and tricycles, where scholars have long noted inefficiencies in routing and fare systems. Older studies, like Santos (2013), analyzed Metro Manila's fragmented jeepney networks causing congestion and poor integration with rail systems, recommending route restructuring without digital solutions [3]. Similarly, a 2016 travel survey by Asian Transport Studies highlighted commuter reliance on informal knowledge for multimodal trips, exposing gaps in accessible planning tools [4].

A 2023 PMC study examined EDSA busway reforms amid COVID-19, accelerating PUV changes but revealing routes were unclear and fares are not transparent or well-known in secondary vehicles like tricycles [5]. Recent works critique the PUV Modernization Program's implementation, noting strains on drivers and commuter displacement without fare transparency aids [6]. These findings underscore a gap: while apps exist for general navigation, none integrate Metro Manila-specific jeepney routes, tricycle zones, and commuter-derived fare estimates—precisely what this study addresses [14].

This topic demands immediate attention amid fuel price surges which have severe and rapid fare changes and route disruptions for Metro Manila commuters[7]. While PUVMP studies focus on fleet upgrades, they overlook micro-level transparency tools for the final part of the commute on tricycles, leaving commuters vulnerable during peak hours. This paper bridges that void with a Metro Manila-specific mobile system using commuter-sourced data, offering practical fare guidance amid rising fare hikes [8][9].

By synthesizing PUVMP fleet upgrade studies, tricycle driver and passenger condition analyses, and fare confusion reports, this research proposes a localized mobile app that features Metro Manila jeepney routes, tricycle zone maps, and commuter-validated fare estimates. The app provides clear and accessible information to help prevent overcharging and addresses the gap left by large-scale transportation reforms, which often neglect small but important details such as exact fares and short routes critical to the final part of the commuters. [8], [15].

A study on ByaHero, a GPS-based app for navigating public transport in Iloilo City, serves as proof of concept for this proposed solution, with 92.6% of surveyed commuters agreeing it eases daily travel by clarifying routes and fares. Its high usability ratings—77.8% overall usefulness and strong demand for features like jeepney route maps (70.4%) and real-time GPS—demonstrate that localized apps effectively address fare confusion and navigation gaps in Philippine public transport systems. Integrating similar validated elements into a Metro Manila-focused tool would thus bridge the "last-mile" challenges overlooked by PUVMP reforms, empowering commuters with reliable, on-demand information. [16].

[1]This work was supported in part by the De La Salle-College of Saint Benilde

Existing research on PUV transport challenges reveals consistent findings across studies: tricycle fares lack standardization, which leads to overcharging, and the PUVMP focuses on fleet upgrades without addressing commuter information needs [8] [12], [11]. This creates a conflict between large-scale modernization policies and local commuting realities, while gaps remain in commuter-focused fare transparency tools for Metro Manila. This study addresses these gaps through a mobile app that integrates survey-validated routes, zones, and fares, directly responding to the disconnect between policy priorities and daily commuter needs.

Statement of the Problem

Commuting in Metro Manila is becoming increasingly difficult, especially because jeepney routes and tricycle zones are often unclear, causing confusion and uncertainty about the correct fare.

The main research question of this study is: *How can we help commuters in Metro Manila identify the correct routes, understand tricycle zones, and estimate fares, thereby reducing confusion and possible overcharging?*

This study seeks to directly address this gap by not merely describing the commuting problem, but by developing a practical route and fare information system that can provide clearer guidance for commuters. By making route and fare information more accessible, this research contributes a solution that improves commuter awareness, supports fare transparency, and helps commuters make better travel decisions, thereby offering a concrete way to make daily commuting more efficient and fair [10] [13].

Scope and Delimitation

Scope:

- Focuses on commuters in Metro Manila.
- Focuses on jeepney and tricycle drivers that are stationed in the area of Metro Manila.
- Examines difficulties in identifying jeepney routes, tricycle zones, and expected transportation fares.
- Collects data through survey questionnaires from commuter respondents.
- Identifies common commuting problems and challenges.
- Aims to improve fare transparency, route awareness, and the overall commuting experience for commuters in general.

Delimitation:

- Limited to commuter respondents only
- Covers only the Metro Manila area and nearby routes.
- Includes only jeepneys and tricycles as modes of transportation.
- Excludes private vehicles (cars, motorcycles), ride-hailing services (Grab, taxis, Angkas, etc.), and train stations (LRT, MRT).
- Does not provide real-time tracking or GPS functionality and focuses on static route/zone/fare info.
- Does not include real-time traffic analysis or predictions.
- Does not guarantee exact fare accuracy (fares are estimates only).
- Does not enforce or regulate official transportation fares.
- Does not include driver/operator perspectives as primary data.
- Relies solely on survey responses, which may include subjective opinions.

Significance of the Study

This study is significant because it addresses a common transportation concern experienced by commuters in Metro Manila: difficulty in identifying jeepney routes, understanding tricycle zones, and estimating fare transportation costs. In areas where commuting information is unclear, commuters may waste time, spend more money than necessary, or become dependent on asking drivers and other people for directions and fare details. This can create confusion and may also lead to fare transparency issues, especially in tricycle trips where overpricing may occur.

The findings of this study will be beneficial to commuters, as they will gain a better understanding of available routes and expected fares, allowing them to make more informed and cost-efficient travel decisions. It may also benefit other commuters who travel through the same area by providing clearer commute information.

For transportation stakeholders such as commuters, jeepney and tricycle drivers, the study may serve as a reference in improving route awareness and fare transparency. It may also help commuters where they frequently experience confusion.

In the field of Information Systems, this study contributes to the growing body of knowledge on how digital tools can support public transportation navigation and fare estimation. It shows how a mobile-based information system can be used to solve a real-world commuting problem through accessible and practical technology. This study is worth investigating because it seeks to improve commuting efficiency, reduce uncertainty, and promote fairness in everyday transportation.

II. METHOD

A. Research Design

The research design used in this study is an exploratory sequential mixed-methods design. This approach first examines the challenges faced by jeepney and tricycle drivers and then identifies patterns among commuters. The goal is to gain a clearer understanding of issues related to route navigation, fare awareness, and the possible use of a mobile-based information system.

In the first phase, qualitative interviews were conducted

with jeepney and tricycle drivers to gather insights into common commuter difficulties in identifying routes, terminals, and fares, as well as their perspectives on the usefulness of the proposed system.

In the second phase, a survey was given to commuters to assess their knowledge of routes, terminals, and fare prices, and to determine how useful they believe the proposed mobile application would be.

The integration of qualitative and quantitative data provides a more comprehensive understanding of commuting challenges and supports the development of an effective, user-centered route and fare information system.

B. Population and Sample

The population of this study includes commuters in Metro Manila, particularly students, persons with disability, and senior citizens who regularly use public transportation, as well as jeepney and tricycle drivers operating in the same area. This population was selected because they represent the primary users and stakeholders of public transportation systems and are directly affected by issues related to route accessibility and fare information.

The study focuses on commuters who regularly rely on public transportation, particularly those without access to private vehicles and those who are not fully familiar with jeepney routes. This group is prioritized as they are the primary beneficiaries of the proposed commuter route and fare information system.

A non-probability purposive sampling method was employed to select respondents who possess relevant experience in commuting or operating public transportation within Metro Manila, Eligible participants include:

1. Commuters who frequently commute using jeepneys or tricycles.
2. Jeepney and tricycle drivers currently operating in the Metro Manila area.

A total of 117 respondents participated in the study, consisting of 101 commuters and 16 drivers. The commuter respondents were selected and surveyed to assess their awareness of transportation routes and fares, as well as their perceived usefulness of a mobile based information system. Meanwhile, the driver respondents, composed of 8 jeepney drivers and 8 tricycle drivers, were interviewed to provide accurate practical insights regarding routes, fare structure, and current transportation practices.

C. Respondent/Participant

The respondents of this study will consist of commuters, jeepney drivers, and tricycle drivers in Metro Manila. The commuter respondents will specifically include those who regularly rely on public transportation, such as Students, Persons with disability, Senior Citizens, those without access to private vehicles, and those who are not fully familiar with jeepney routes. These groups are selected because they are the primary users and stakeholders of the proposed commuter route and fare information system. Their participation is expected to provide valuable insights into common commuting challenges, route accessibility, and levels of fare awareness, which are essentials for the development and improvement of the system.

The study will include approximately 117 respondents, divided into three groups:

- 101 commuters who regularly commute using jeepneys and tricycles.
- 8 jeepney drivers operating along Metro Manila routes.
- 8 tricycle drivers serving Metro Manila.

The commuters are chosen to represent the commuter or end-users of the system, while the jeepney and tricycle drivers are included to provide accurate information and insights into current transportation practices.

A purposive sampling method will be used in selecting respondents, ensuring that only individuals who have relevant experience with commuting or drivers within Metro Manila are included in the study.

D. Instruments

The two primary data collection instruments used in this study were a semi-structured interview (Phase 1) and an online survey (Phase 2). For the first phase, the semi-structured interview was designed to explore the difficulties commuters face when it comes to identifying routes, terminals, and fares. It is also used to provide insights, from the driver's perspective, on the perceived usefulness of a mobile application that can guide commuters. For the second phase, online surveys were used to measure the commuters' awareness of the fares, routes, and terminals. It was also used to gain the perceived usefulness of the mobile application, but in the perspective of the commuters.

Interview Design

Interview questions were translated into Filipino to ensure clarity and enhance comprehension among the participants, thereby improving the reliability of their responses.

1. Respondent Details

   This section is used for gathering background information of the participants (i.e. Jeepney and Tricycle Drivers). This includes their:

   - Age
   - Name
   - How long have they been doing their profession

- What kind of driver they are (Jeepney Driver or Tricycle Driver)

2. Community, Route, and Fare Decision-Making

This section explores how the driver's community, such as a TODA or cooperative, influences decisions regarding fares and routes. It also examines how routes are assigned, how fare rules are followed, and whether group agreements affect daily transport operations. These questions help identify the role of People and Process in the transportation system.

3. Passenger Behavior and Fare Interactions

This section examines how passenger behavior affects the driver's mood, driving style, and interactions during the trip. It also looks at how drivers handle fare complaints, haggling, or misunderstandings, and whether passenger attitude influences the overall commuting experience. This section supports the study's focus on the People component of the Information Systems framework.

4. Standard Fare Procedure and Fare Basis

This section investigates the usual process drivers follow when picking up passengers and setting fares. It includes questions on whether fares are based on distance, boundary rules, route sections, or official fare matrices. This helps identify how fare decisions are made in actual practice and whether the procedure is clear and consistent.

5. Official Fare Updates and Transparency

This section examines where drivers get official fare updates, such as LTFRB announcements, LGU signage, cooperatives, or transport groups. It also asks whether fares are displayed or explained to passengers and why this may or may not happen regularly. These questions help determine if fare transparency is affected by the lack of clear information or signage.

6. Fare Overcharging and Root Causes

This section explores situations where drivers may unintentionally charge more than the standard fare and identifies possible causes such as confusion, unclear rules, passenger requests, or boundary-related misunderstandings. It also investigates whether rising fuel and maintenance costs affect pricing decisions. This section is important in identifying whether the problem is behavioral, procedural, or technical.

7. Barriers to Sharing Fare Information

This section looks into the reasons why drivers cannot always explain or display fare information upfront. These may include limited time, traffic, lack of signage, and the busy nature of public transport operations. This helps explain why commuters may remain unaware of the correct fare even when fare rules exist.

8. Perspectives on the Proposed Solution

This section gathers the drivers' views on the proposed mobile application that provides route guidance, tricycle zones, and estimated fares. It also checks whether a digital fare reference could reduce misunderstandings, increase trust, improve passenger flow, and lessen conflict between drivers and commuters. This section supports the Technology component of the Information Systems framework.

Survey Design

1. Screening

This section is used to screen respondents and ensure that only commuters who regularly travel within Metro Manila are included in the study. It identifies their usual commuting area, mode of transportation, and frequency of commuting.

2. Demographic

This section is used for gathering background information of the respondents. This includes:

- The city they reside inside Metro Manila
- Are they a Student, Person with Disability, or Senior Citizen

3. Route and Zone Confusion

This section aims to assess the level of commuter awareness and understanding of jeepney routes and tricycle zones within Metro Manila. It identifies how often commuters experience confusion when selecting the correct transportation option and whether this confusion affects their travel decisions. The data gathered from this section will help determine whether the problem is due to lack of knowledge, unclear transportation systems, or procedural issues, which are essential in identifying the root causes of commuting difficulties.

4. Fare and Overcharging

This section aims to assess commuters' awareness of fare rates and their experiences related to fare transparency and possible overcharging within Metro Manila. It examines whether commuters know the correct fare for their usual trips, how often they experience fare

uncertainty, and whether they have encountered situations where they were charged more than expected. The data gathered from this section will help determine the extent of fare-related problems and support the need for a route and fare information system that can improve transparency and reduce confusion.

5. Root Cause

This section aims to identify the underlying reasons why commuters experience route confusion, fare uncertainty, and possible overcharging in Metro Manila. It examines whether these problems are mainly caused by commuter behavior, driver behavior, unclear procedures, lack of accessible information, or the absence of supporting technology. The data gathered from this section will help determine the root causes of commuting difficulties and support the development of a more effective and user-centered information system.

6. Mobile App Value

This section aims to determine whether commuters see the proposed mobile application as useful in addressing route confusion, fare uncertainty, and overcharging concerns in Metro Manila. It examines whether commuters would use the app, how it compares to asking drivers or other passengers, and what benefits they expect from it such as faster access to information, improved convenience, and better decision-making. The data gathered from this section will help establish whether the proposed system has practical value for commuters and whether it can serve as a helpful information tool rather than just a fare database.

7. Feedback System

This section aims to evaluate the usefulness of integrating a feedback feature within the proposed mobile application. It focuses on determining whether commuters are willing to report issues such as overcharging, incorrect fare information, or route inaccuracies. It also examines how likely users are to actively participate in providing feedback and what type of feedback they are willing to share. The data gathered from this section will help assess whether a feedback mechanism can improve data accuracy and contribute to reducing fare-related conflicts between commuters and drivers.

8. Information System Framework (People, Process, Technology)

This section evaluates commuting problems using the Information Systems framework by examining the role of people, process, and technology. It seeks to identify the main cause of overcharging, determine whether a mobile application can improve fare transparency, and assess whether better information can reduce commuter stress. The responses from this section will help explain whether the proposed system addresses the problem through user behavior, clearer procedures, and digital support.

E. Data Gathering Procedure

The data gathering procedure for this study follows an exploratory sequential approach, where there are two phases. The first phase is a semi-structured interview meant to gather information from the drivers and the second phase is a digital survey that is distributed amongst commuters to gather data on their awareness of routes and fares within the area of Metro Manila. The data collection for the semi-structured interview was conducted face-to-face and the digital survey was created using Google Forms and distributed through online platforms.

Recruitment and Sampling

Participants were recruited using purposive non-probability sampling to ensure individuals had direct experience with Metro Manila commuting using jeepneys or tricycles. For the 101 commuters, recruitment occurred through a targeted instant messaging application (Messenger App). The 16 drivers (8 jeepney, 8 tricycle) were approached face-to-face at the loading zones. This approach secured 117 total respondents.

Data Collection Time Frame

Data collection followed the exploratory sequential design. (Phase 1) semi-structured interviews with the 16 drivers (8 jeepney, 8 tricycle) were conducted face-to-face. The data collection period started on March 29 and lasted until April 5. Each lasting 15-30 minutes per participant.

(Phase 2) commuter survey (101 respondents) opened on April 1 through Google Forms and closed on April 18, allowing 18 days for responses through online distribution.

Informed Consent and Confidentiality

All participants received a clear informed consent form detailing the study's purpose (commute route and fare issues in Metro Manila), procedures (interviews and surveys), voluntary participation, and confidentiality protections before data collection began. Participation was voluntary; respondents could withdraw anytime without penalty, including during interviews or before survey submission.

Data Security and Quality Assurance

Raw data from driver interviews was audio-recorded with consent and stored in password-protected Google Drive folders and survey responses remained within Google Forms

strictly accessible only to the research team (Gaspar, Garcia, Lopez, Delos Santos).

Ethical Considerations

The study adhered to ethical research principles on DLS-CSB's institutional guidelines for student research. Participants provided informed consent with full disclosure of study aims, methods, and rights; no deception was employed. Minimal risks (primarily time expenditure of 2-20 minutes) were mitigated through flexible scheduling for drivers during non-peak hours and brief survey design. No sensitive personal data (e.g., financial details, health status) was collected beyond basic demographics required for commuter profiling. Raw data access remained restricted to the research team, with no third-party sharing. Conflicts of interest were absent; all researchers maintained objectivity. Findings will be reported accurately, acknowledging limitations like self-reported biases.

F. Statistical Treatment of Data and Data Analysis

Using thematic analysis for the interviews conducted with jeepney and tricycle drivers, the collected data will be systematically extracted, coded, and transformed into a structured qualitative form. This process involves identifying recurring patterns, themes, and key insights that emerge from the participants' responses. The transformed data will be stored and organized using Google Forms, where each response will be encoded according to its corresponding question to ensure clarity and consistency. To maintain the integrity and reliability of the data, measures such as peer review and thorough double-checking were implemented, ensuring that all recorded information remains accurate, coherent, and easily interpretable.

For the commuter surveys, descriptive statistics will be employed to analyze and summarize the data, allowing for the identification of general trends, patterns, and perceptions among respondents. Google Forms will also serve as the primary tool for organizing and visualizing the collected data, as it was both the platform used for survey distribution and response collection. Upon the closure of the survey, each individual response was carefully reviewed to eliminate any irrelevant or inconsistent entries, thereby ensuring the overall accuracy, completeness, and reliability of the dataset.

III. Results and Discussion

This chapter presents and discusses the findings gathered from the survey and interviews conducted in this study in Metro Manila. It focuses on the experiences of commuters, especially the challenges they face in identifying jeepney routes, understanding tricycle zones, and knowing the correct fares. The results provide a clear view of the common problems encountered in daily commuting and how these affect convenience, cost, and decision making.

The data collected from respondents are analyzed to identify key issues such as confusion in routes, lack of awareness of fare rates, and inconsistent pricing, particularly in tricycle services. The responses from both commuters and drivers are used together to explain the findings, giving a more complete understanding of the current transportation situation. These results also show that many commuters still rely on asking drivers or other passengers for information instead of using digital tools.

The analysis highlights how factors such as commuting frequency, experience, and familiarity with transportation systems influence the difficulties faced by commuters. These findings help identify what commuters need, particularly the demand for clear, accessible, and reliable information. Overall, the results serve as a basis for the proposed mobile application, which aims to improve route guidance, provide fare estimates, and make commuting easier and more efficient.

G. Data Presentation and Analysis

Figure 1 presents the distribution of respondents by city within Metro Manila. The majority of respondents are concentrated in Parañaque (11:2%), Taguig (8.8%), and Caloocan (8.8%), indicating that the dataset is largely derived from respondents in these areas..

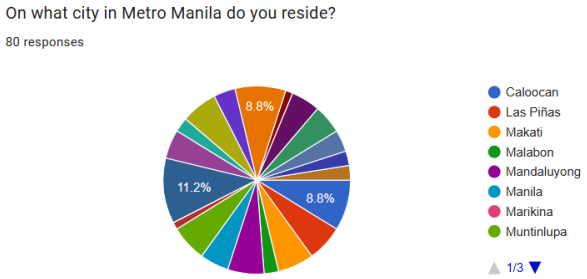


Fig. 1. Distribution of respondents by city of residence in Metro Manila

Figure 2 presents the percentage of respondents who are students, persons with disabilities, or senior citizens. The majority of respondents (62.5%) answered "Yes," indicating that a large portion of the dataset consists of individuals belonging to these groups..

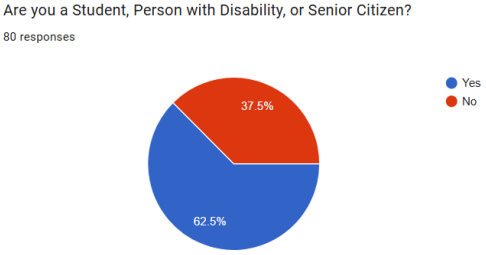


Fig. 2. Percentage of respondents who are students, persons with disability, or senior citizens

Figure 3 illustrates the areas to which respondents commonly commute to. The majority of respondents reported traveling to Manila (62.5%), Makati (45%), and Pasay (38.8%), suggesting that these locations are the primary commuting destinations within the dataset.

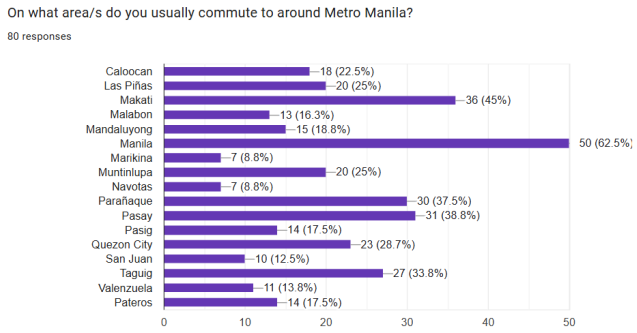


Fig. 3. Areas in Metro Manila where respondents commonly commute

Figure 4 shows the modes of transportation used by respondents. The majority (53.8%) reported using both jeepneys and tricycles for their commute, while 42.5% rely solely on jeepneys. A small portion (3.7%) indicated that they use only tricycles. This suggests that multimodal transport is common among the respondents.

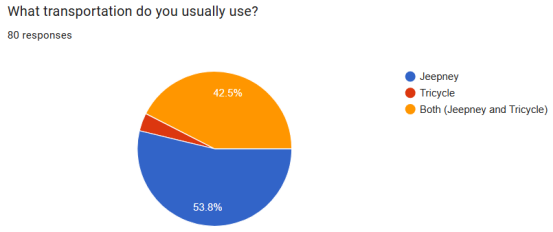


Fig. 4. Primary mode of transportation used by respondents

Figure 5 presents the commuting frequency of respondents in Metro Manila. The majority reported commuting several times a week (65%), followed by those who commute daily (23.8%). A smaller proportion indicated commuting once a week (8.8%) or a few times a month (2.5%), and none reported commuting rarely, suggesting generally high travel frequency among the sample.

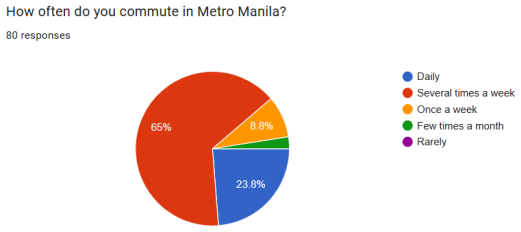


Fig. 5. Respondents' commuting frequency in Metro Manila

Figure 6 illustrates the respondents' confidence in identifying correct jeepney routes, measured on a 5-point Likert scale (1 = very low confidence, 5 = very high confidence). Most respondents reported a neutral level of confidence (46.3%). A considerable share expressed low confidence (25% low and 7% very low), while a smaller proportion reported high confidence (11.3% high and 8.8% very high), This distribution suggests that many respondents are uncertain about their ability to identify correct jeepney routes.

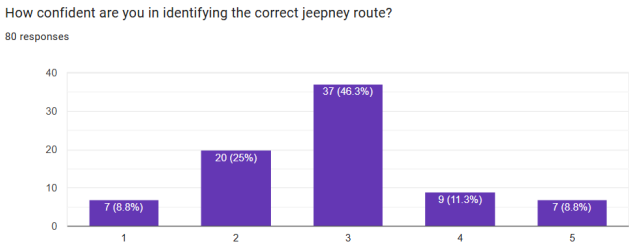


Fig. 6. Respondents' confidence level in identifying correct jeepney routes

Figure 7 shows the respondents' confidence level in identifying correct tricycle routes, measured on a 5-point Likert scale (1 = very low confidence, 5 = very high confidence). Similar to Figure 6, the largest proportion of respondents reported a neutral level of confidence (43.8%). A considerable share expressed low confidence, with 22.5% indicating low confidence and 10% very low confidence. In contrast, fewer respondents reported high confidence, with 11.3% indicating high confidence and 8.8% very high confidence. This distribution suggests that many respondents remain uncertain about their ability to identify correct tricycle routes.

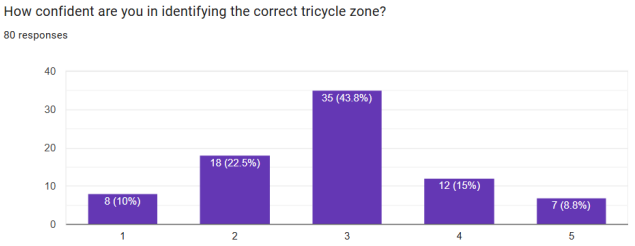


Fig. 7. Respondents' confidence level in identifying correct tricycle routes

Figure 8 shows whether respondents have experienced taking the wrong route in Metro Manila. The majority (93.8%) answered "Yes," indicating that most respondents have encountered this issue when travelling to their destination.

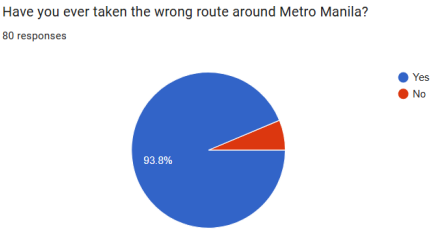


Fig. 8. Respondents who have experienced taking the wrong route in Metro Manila

Figure 9 presents the frequency with which respondents feel confused about which vehicle to ride. Responses were measured using a 5-point frequency rating scale (Never = Lowest, Rarely = Low, Sometimes = Neutral, Often = High, Always = Highest). The majority of respondents reported feeling confused sometimes (40%) or often (40%). A smaller proportion indicated that they rarely (8.8%) or always (8.8%) experience confusion, while only 2.5% reported never feeling confused.

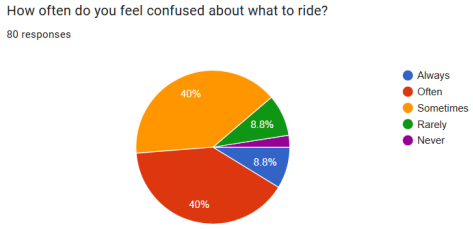


Fig. 9. Frequency of respondents feeling confused about which vehicle to ride

Figure 10 illustrates respondents' awareness of correct fare calculations in Metro Manila. A large majority (77.5%) reported a lack of awareness, while only 22.5% indicated familiarity with correct fares, suggesting that fare knowledge is limited among respondents.

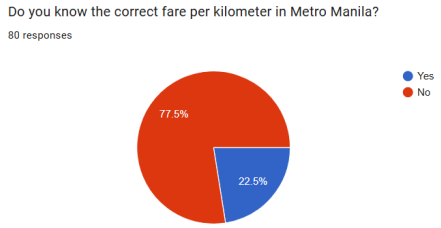


Fig. 10. Respondents' awareness of the correct fare per kilometer in Metro Manila

Figure 11 presents the frequency of respondents' uncertainty about fares in Metro Manila, measured using a 5-point frequency rate scale (Never = Lowest, Rarely = Low, Sometimes = Neutral, Often = High, Always = Highest). The majority of respondents reported often feeling uncertain (42.5%), followed by those who sometimes feel uncertain (33.8%) and always feel uncertain (13.8%). Smaller proportions indicated rarely (7.5%) or never (2.5%) experiencing uncertainty. These results suggest that respondents frequently encounter uncertainty regarding fare costs.

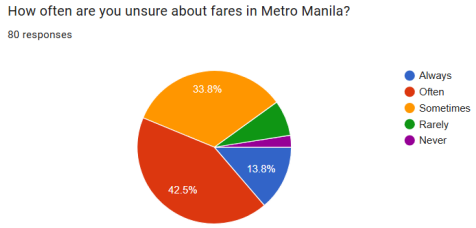


Fig. 11. Frequency of respondents' uncertainty about fares in Metro Manila

Figure 12 illustrates the proportion of respondents who have experienced being overcharged. Respondents selected from four options: "Yes, often," "Yes, sometimes," "Yes, rarely," and "No, never." The majority reported experiencing overcharging sometimes (48.8%), followed by those who experienced it often (22.5%) and rarely (20%). Only 8.8% indicated that they have never been overcharged, suggesting that this issue is widespread among the respondents.

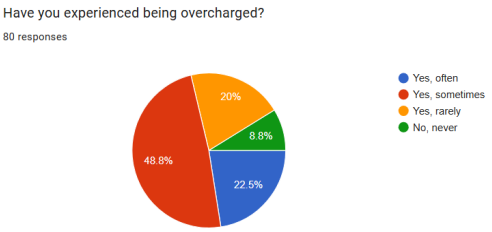


Fig. 12. Proportion of respondents who experienced overcharging

Figure 13 shows the actions taken by respondents when they are unsure of the fare. The item aims to identify how respondents obtain fare-related information. The most common actions were asking the driver (78.8%) and asking a friend or relative (76.3%). Other respondents reported asking fellow passengers (55%) or checking online sources (43.8%), while only a small portion asked a co-worker (1.3%).

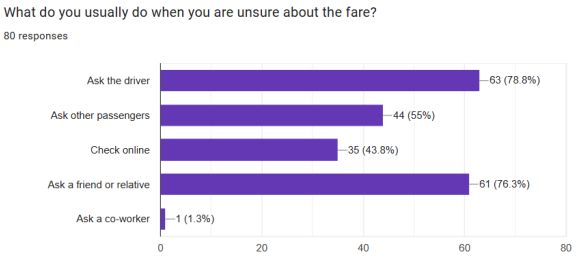


Fig. 13. Actions taken by respondents when unsure about fares

Figure 14 displays the actions taken by respondents when they are overcharged. The majority reported that they still pay the fare (47.5%) or do nothing (13.7%). Others indicated that they either negotiate or argue with the driver (27.5%), while 10% reported asking fellow passengers about the correct fare, suggesting limited active resolution strategies among respondents.

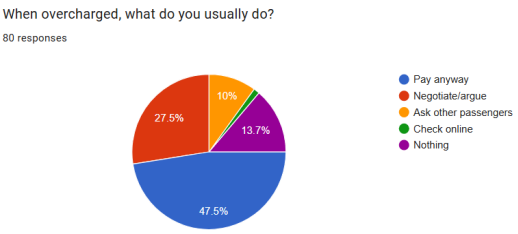


Fig. 14. Actions taken by respondents when overcharged

Figure 15 presents the level of trust respondents have in drivers regarding fare accuracy. The responses are relatively evenly distributed, with the highest proportion reporting high trust (27.5%), followed by low trust (26.2%), neutral (21.3%), and very low (18.8%). Only a small proportion of respondents indicated very high (6.3%) trust in drivers, suggesting mixed perceptions of driver reliability in fare determination.

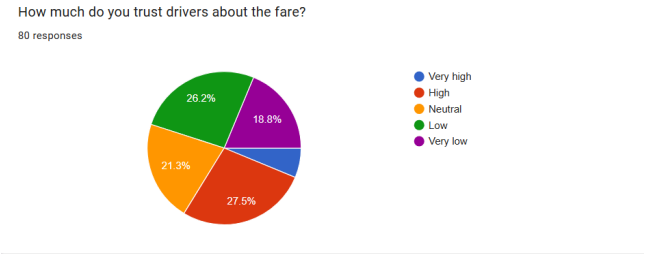


Fig. 15. Level of trust of respondents in drivers regarding fare accuracy

Figure 16 presents the main reasons contributing to commuting confusion among respondents. The majority identified lack of information (88.8%) and unclear rules or procedures (63.7%) as key causes. Other contributing factors include inconsistent driver pricing (55%) and the absence of signs or maps (41.3%), suggesting that both informational gaps and system inconsistencies play a significant role in commuter confusion.

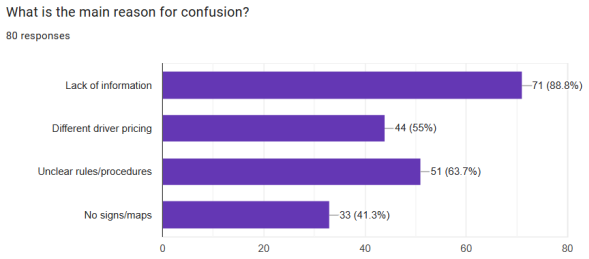


Fig. 16. Main reasons for commuting confusion among respondents

Figure 17 presents the perceived causes of overcharging among respondents. The majority identified unclear pricing systems (80%) and lack of information (77.5%) as the primary contributors. Other factors include driver behavior (53.8%) and the absence of a mechanism to verify fares (41.3%), indicating that both systemic issues and individual practices contribute to overcharging.

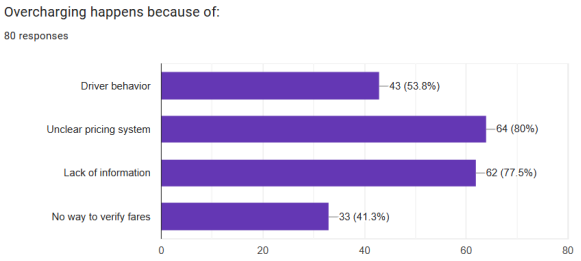


Fig. 17. Perceived causes of overcharging among respondents

Figure 18 presents the respondents' perceptions of whether driver procedures contribute to commuting issues, measured on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Most respondents expressed agreement, with 35% agreeing and 30% strongly agreeing that unclear procedures lead to problems. Meanwhile, 23.8% remained neutral, indicating some variation in perceptions but an overall tendency towards agreement.

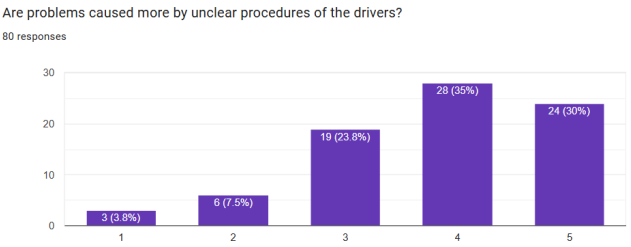


Fig. 18. Respondents' view on whether driver procedures cause commuting issues

Figure 19 displays respondents' willingness to use a mobile commuting guide application. Most respondents expressed strong interest, with 63.7% indicating they would definitely use the application and 27.5% stating they would probably use it. Only a small proportion (8.8%) were uncertain, suggesting generally high acceptance of the proposed solution.

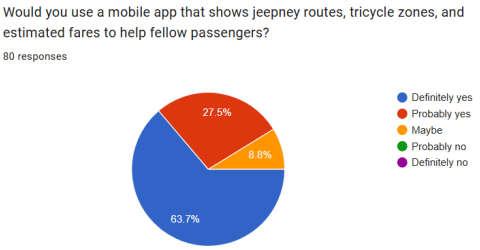


Fig. 19. Willingness of respondents to use a mobile commuting guide application

Figure 20 shows respondents' perceptions of the time efficiency of using the proposed commuting guide application compared to asking drivers. A large majority (82.5%) perceived the application as faster, while 11.3% indicated no difference and only 6.3% believed it would be slower. These findings suggest strong perceived efficiency benefits of the application.

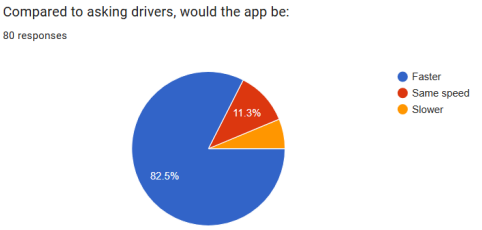


Fig. 20. Perceived time advantage of using a mobile app over asking drivers

Figure 21 shows the respondents' perceptions of the reliability of the proposed mobile application compared to asking drivers. Most respondents (83.8%) perceived the application as more reliable, while 13.8% indicated no difference and only 2.5 viewed it as less reliable. These findings suggest strong confidence in the application's reliability as a source of fare and commuting information.

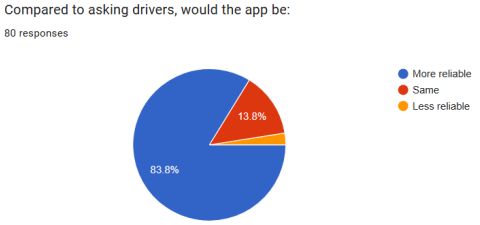


Fig. 21. Perceived reliability advantage of a mobile app over asking drivers

Figure 22 shows respondents' perceptions of the value of downloading the proposed mobile application. An overwhelming majority (97.5%) agreed that the application would be worth downloading, indicating strong potential user acceptance.

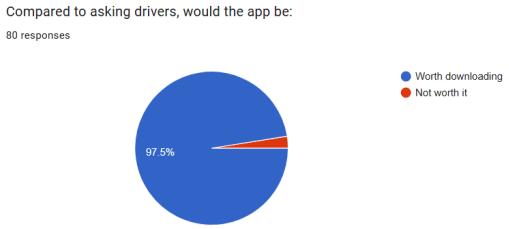


Fig. 22. Perceived worthiness of downloading the mobile app

Figure 23 shows respondents' perceptions of the time-saving potential of the application. The largest group (41.3%) believed that it would save more than 10 minutes, while 26.3% indicated no expected time savings. Others estimated more modest benefits, with 21.2% reporting 1-3 minutes, and smaller proportions indicating 7-10 minutes (6.3%) and 4-6 minutes (5%). These results suggest varied expectations, though a substantial portion anticipates significant time savings.

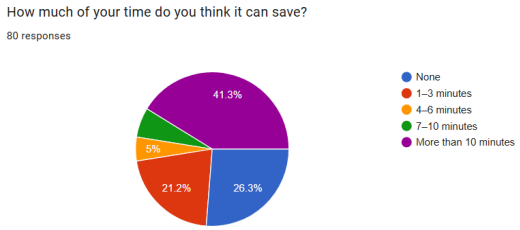


Fig. 23. Respondents' perception of time saved by the application

Figure 24 illustrates respondents' perceptions of potential monetary savings from using the mobile application. The largest group (37.5%) indicated that no savings would be achieved, while a substantial proportion (31.3%) believed they could save more than ₱20. The rest of the respondents anticipated smaller savings within the ranges of ₱1-₱20, suggesting mixed expectations regarding the application's financial benefits.

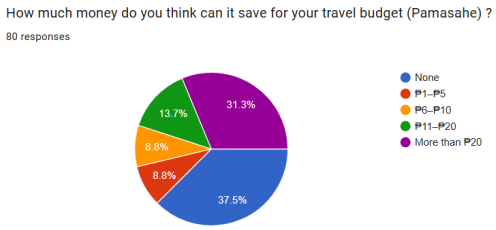


Fig. 24. Respondents' perception of money saved from using the mobile app

Figure 25 displays respondents' willingness to report overpricing through the mobile application. Most respondents (72.5%) expressed willingness to report, while 22.5% were uncertain and 5% indicated that they would not report. These results suggest generally positive user intent toward reporting fare-related issues.

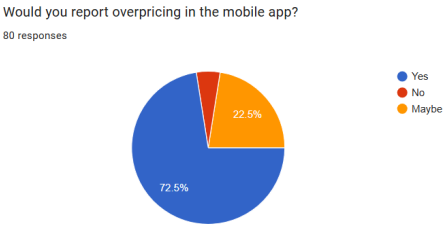


Fig. 25. Willingness of respondents to report overpricing via the mobile app

Figure 26 illustrates respondents' willingness to provide feedback through the mobile application, measured on a 5-point Likert scale (1 = Very unlikely, 5 = Very likely). Most respondents (61.3%) reported that they are very likely to provide feedback, followed by 15% who are likely to do so. Meanwhile, 12.5% were uncertain, and only a small proportion indicated unlikely (6.3%) or very unlikely (5%) responses. These results suggest a strong overall intention to engage with the feedback feature.

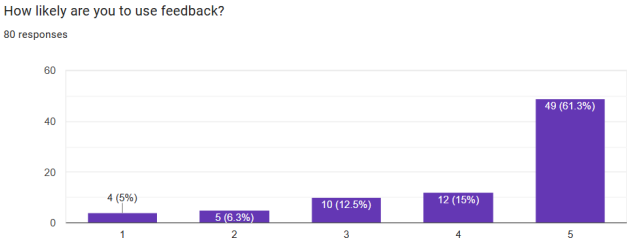


Fig. 26. Willingness of respondents to provide feedback via the app

Figure 27 presents respondents' perceptions of the main causes of overcharging. Most respondents identified a poor system or process as the primary cause (42.5%), while other respondents believed that commuters' lack of knowledge (18.8%), driver behavior (12.5%), and lack of technology (6.3%). Additionally, 20% believed that all listed factors contribute equally. These findings suggest that overcharging is perceived as a multi-factor issue, with systemic problems being the most prominent.

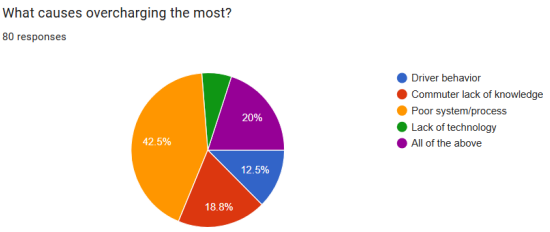


Fig. 27. Respondents' perception of the main cause of overcharging

Figure 28 illustrates respondents' perceptions of the potential of a mobile application to improve fare transparency, measured on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Most respondents expressed strong agreement (60%), followed by agreement (22.5%). Meanwhile, 11.3% remained neutral, while only a small proportion disagreed (3.8%) or strongly disagreed (2.5%). These results indicate a generally positive perception of the application's potential to enhance face transparency.

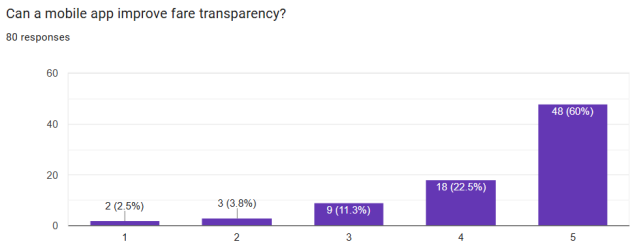


Fig. 28. Respondents' view on whether a mobile app can improve fare transparency

Figure 29 shows the respondents' view on whether better information can reduce stress, measured on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Most respondents expressed strong agreement (65%), followed by agreement (23.8%). Meanwhile, 10% remained neutral, while only a small proportion disagreed (1.3%). These results indicate a generally positive perception of better information to reduce stress among respondents.

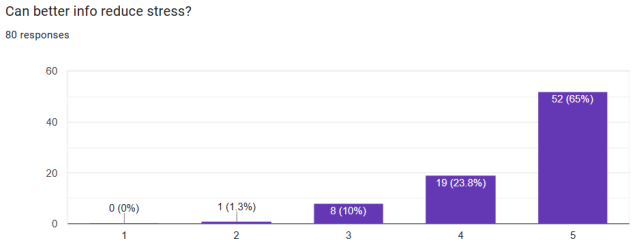


Fig. 29. Respondents' view on whether better information can reduce stress

Thematic Analysis of the semi-structured interviews

This section presents the thematic analysis of the data gathered from jeepney and tricycle drivers. The aim of this analysis is to identify recurring patterns and themes that address the research question, which is "*How can we help commuters in Metro Manila identify the correct routes, understand tricycle zones, and estimate fares, thereby reducing confusion and possible overcharging?"*

Theme 1: Institutional Regulation through TODA and Cooperative

Findings show that fare rates and route decisions are largely controlled by TODA organizations and cooperatives rather than individual drivers. Respondents consistently stated that they follow group decisions regarding fare implementation, route assignments, and queue systems. These institutions serve as the main enforcers of discipline and coordination, ensuring compliance with LTFRB guidelines. Drivers noted that violations of TODA rules may result in penalties or restricted access to Terminals

Theme 2: Fare Determination Based on Regulation and Operating Conditions

The data indicates that fare computation is primarily based on LTFRB fare matrices and travel distance. However, drivers also consider operational conditions such as fuel prices, boundary payments, traffic congestion, and trip type (regular vs. special trips). While standard fares are followed for regular routes, adjustments in practice occur depending on economic and situational factors affecting daily income.

Theme 3:Passenger Behavior and Emotional Response

All participants reported that passenger behavior influenced their mood and driving behavior, though not the official fare. Positive passenger interactions (e.g., polite communication and exact payment) contribute to smoother operations, while negative behavior (e.g., complaints and rudeness) increases stress and reduces patience. Despite this, drivers generally maintain professionalism due to fear of complaints, reporting, or public exposure.

Theme 4: Limited Fare Awareness Among Commuters

Drivers consistently observed that many commuters lack accurate or updated knowledge of fare rates. This is attributed to outdated or missing fare matrices, limited signage visibility, irregular updates from authorities, and reliance on informal information sources. Misunderstandings commonly arise when passengers assume outdated fares or are unfamiliar with specific route-based pricing systems.

Theme 5: Economic Pressure and Operational Constraints

A dominant theme across all interviews is financial pressure. Respondents identified rising fuel costs, daily boundary obligations, vehicle maintenance expenses, and fluctuating passenger volume as major challenges. These factors directly affect income stability and influence operational decisions such as working hours, trip frequency, and route efficiency.

Theme 6: Perceived Value of Digital Fare Information Systems

Most participants expressed support for digital fare reference systems such as mobile applications, QR codes, or digital fare displays. They believe such systems could reduce fare disputes, improve transparency, and speed up boarding processes. However, concerns were raised regarding accessibility for non-tech-savvy passengers, internet reliability, and the need for consistent updates.

H. Discussion of Results

The findings of the study show that the problem of route confusion, fare uncertainty, and overcharging is experienced across multiple areas in Metro Manila. The respondent profile already supports this broader scope, since commuters came from different cities and commonly traveled to major destinations such as Manila, Makati, and Pasay. This confirms that the issue is citywide and justifies expanding the study to a Metro Manila-wide context. The results suggest that fare-related problems are shaped by the commuting environment as a whole.

The survey data further show that commuters often struggle with identifying the correct route and knowing the proper fare. Many respondents reported low or neutral

confidence in identifying jeepney and tricycle routes, and most of them had already experienced taking the wrong route. In terms of fare knowledge, a large majority said they were not aware of the correct fare rate, and many also reported that they often feel uncertain about how much they should pay. These findings answer the research problem by showing that route and fare information remains unclear for many commuters, which leads to confusion and increases the risk of overcharging.

The interview results reveal that the issue is not purely informational. It is also affected by the People and Process pillars of Information Systems. On the People side, drivers explained that passenger behavior affects their mood and interactions during trips, while commuters often depend on drivers or other passengers for fare information instead of having a reliable reference. On the Process side, fare computation and route decisions are influenced by TODA or cooperative rules, official fare matrices, boundary systems, traffic conditions, and fuel costs. This means that overcharging is not caused by one factor only, but by the interaction of behavior, procedure, and environment. The results therefore support the study should explicitly consider root causes as behavioral, procedural, and technical.

The results also show that tricycle concerns are not confined to a single area. Drivers identified fare implementation issues, lack of commuter awareness, inconsistent updates, and operational pressure as common concerns. These are consistent with the broader Metro Manila setting of the study. In addition, the interviews revealed that drivers often rely on institutional rules through TODA or cooperatives, but commuters still do not always receive clear fare explanations. This gap between formal rules and actual passenger awareness explains why overcharging can still happen even when fare guidelines exist.

The proposed mobile application has clear value because respondents viewed it as more useful than repeatedly asking the driver. Many commuters said the app would be faster, more reliable, and worth downloading. A large majority also believed it could save time, and some expected monetary savings as well. These results prove that the system is not just a static fare matrix database. Instead, it functions as an Information System because it processes and delivers relevant route and fare information in a way that supports decision-making for commuters, drivers, and other stakeholders. The app adds practical value by reducing dependence on guesswork and by making fare information easier to access before boarding.

Another important finding is the strong support for the feedback feature. Most respondents were willing to report overcharging and provide feedback through the app. This is significant because it allows the system to go beyond information display and become a tool for monitoring real fare issues. A feedback mechanism helps identify incorrect fare information, route mismatches, and possible overpricing incidents, which can then be used to improve the accuracy of the system. This directly addresses that the app must help monitor overcharging and not merely list fare data.

Overall, the findings show that the proposed system can address the commuting problem through all three Information Systems pillars. It supports the People pillar by helping commuters make better decisions and helping drivers reduce repeated fare disputes. It supports the Process pillar by making fare information clearer and more consistent. It supports the Technology pillar by providing a digital platform for route guidance, fare reference, and feedback reporting. In this sense, the study answers its research objectives by showing that a commuter route and fare information system can reduce confusion, improve fare transparency, and provide measurable benefits such as time savings, reduced effort, and possible money savings.

The findings align with prior studies on Philippine transport gaps: Santos (2013) noted fragmented jeepney networks without digital aids [3] and tricycle fare non-standardization and overcharging [7] [11]. These insights underscore the app's potential to empower students to cut disputes, and boost transportation efficiency.

I. Theoretical Contribution and Practical Implications

This study contributes to the theoretical understanding of commuter route and fare information systems by showing that overcharging and route confusion in Metro Manila are not caused by fare information alone, but by the interaction of the People, Process, and Technology pillars of Information Systems. The findings extend the literature by moving beyond a simple fare matrix approach and demonstrating that commuter problems are also shaped by driver thinking, passenger behavior, organizational practices such as TODA or cooperative rules, and the lack of a reliable digital reference for fare verification. In this sense, the study supports the idea that an effective Information System must do more than store information; it must help users interpret, verify, and act on that information in real commuting situations.

The study also broadens the theoretical scope of fare transparency research to the wider Metro Manila commuting context. This research is important because the results show that route confusion and fare uncertainty are not isolated to one area, but are experienced across multiple cities and travel destinations. By including both jeepney and tricycle drivers, the study further strengthens the literature on public transport information systems because tricycle fare issues, driver procedures, and commuter uncertainty were shown to be major and recurring concerns. This makes the study more relevant to real Metro Manila commuting conditions and improves its contribution to Information Systems research.

Another theoretical contribution of the study is its identification of the root causes of overcharging as behavioral, procedural, and technical rather than purely informational. The interviews and survey results show that fare disputes are influenced by driver behavior, unclear fare procedures, inconsistent information sharing, and the

absence of a fast and accessible verification tool. This helps explain why a fare information app can be theoretically justified as an Information System: it does not merely display fares, but supports better decision-making, reduces uncertainty, and creates a feedback loop for monitoring incorrect pricing. The addition of a feedback feature strengthens this contribution because it transforms the app from a passive database into a system that can collect reports and support continuous improvement.

In practical terms, the findings suggest that commuters are likely to benefit from a Metro Manila-based mobile application because they see it as faster, more reliable, and more useful than repeatedly asking drivers or relying on informal sources. The respondents also indicated willingness to use the app, save time, and possibly save money and effort. This means the proposed system has real value for commuters in general, jeepney drivers, and tricycle drivers because it can reduce repeated fare disputes, improve fare transparency, and make commuting decisions easier before boarding. The app therefore has practical value not just as a reference tool, but as a commuter support system that can improve daily travel efficiency.

For drivers, the system may also reduce stress caused by fare arguments and repeated explanations, especially when commuters already know the expected fare. For commuters, especially frequent riders, the system can provide a clearer basis for deciding which vehicle to take, how much to pay, and whether a fare seems excessive. For transport groups and local stakeholders, the feedback feature can help identify recurring overpricing concerns and route-related issues, which may guide future fare monitoring and information campaigns. In this way, the study supports practical improvements in communication, transparency, and accountability within Metro Manila public transportation.

Overall, the study shows that the proposed mobile application is not simply a fare matrix database. It is an Information System because it combines route guidance, fare estimation, user awareness, and feedback reporting to address a transportation problem through the People, Process, and Technology framework. These findings directly support the proposed capstone project by showing that the system has theoretical relevance, practical usefulness, and the potential to reduce overcharging through better information access and monitoring.

J. Limitations

This study has several limitations that should be considered in interpreting the results. First, although the scope was expanded to Metro Manila, the data were still gathered through purposive sampling and were limited to respondents who were accessible to the researchers. Because of this, the findings may not fully represent all commuters, jeepney drivers, and tricycle drivers across the entire region. The sample size is still not large enough to capture every commuting experience in Metro Manila, especially across all cities, routes, and transport contexts. As a result, the findings should be understood as indicative rather than fully generalizable.

Another limitation is that the study relies partly on self-reported responses from commuters and drivers. This means the results may be affected by personal perception, memory, and individual experience, especially in questions about overcharging, fare confusion, and route difficulty. Because the study focuses on perceived experiences, it may not fully measure actual fare violations or objectively verify every reported incident. In addition, while the study now considers the People, Process, and Technology components of the Information Systems framework, the root causes of overcharging are still interpreted based on respondent insights rather than direct operational observation of every transport group or terminal procedure.

The study also does not yet include actual system deployment, user testing over time, or live monitoring of how the app performs in real commuting situations. Although the proposed feedback feature addresses overcharging concerns, its effectiveness cannot yet be fully proven until the system is developed and tested in practice. Likewise, the estimated benefits in time, money, and effort are based on respondent perceptions rather than measured before-and-after performance data. Therefore, while the findings support the value of the proposed application, further implementation and evaluation are still needed to confirm its long-term effect on fare transparency and overcharging reduction. Critically, static fare estimates data cannot adapt to rapid fuel-driven hikes or LTFRB route updates, This likely overstates short-term app accuracy reducing generalizability beyond data collection timing.

IV. Conclusion

In conclusion, the study shows that route confusion, fare uncertainty, and overcharging are not isolated problems in one location but are recurring concerns across Metro Manila. The findings confirm that commuters often struggle to identify the correct jeepney or tricycle route, determine the proper fare, and rely on drivers or other passengers for information. At the same time, the interviews with drivers show that fare-related issues are influenced not only by information gaps, but also by the People, Process, and Technology components of the transport environment. Driver behavior, organizational practices such as TODA or cooperative procedures, and the lack of a clear digital reference all contribute to the persistence of overcharging and fare disputes.

Based on the results, the proposed commuter route and fare information system has clear value for commuters, jeepney drivers, and tricycle drivers. Respondents indicated that the system would be useful because it can provide faster access to fare and route information, reduce uncertainty, and help commuters avoid overpaying. The feedback feature also strengthens the system by allowing users to report incorrect fares or overpricing incidents, making the application more responsive and more than just a fare

database. This supports the study's main objective of developing an Information System that can help improve fare transparency and commuting decision-making in Metro Manila.

Overall, the study concludes that the proposed application has the potential to address overcharging by improving access to accurate information, supporting clearer procedures, and encouraging better communication between commuters and drivers. Future research may further improve this study by increasing the sample size, including more transport routes and cities outside Metro Manila, and conducting actual system testing to measure real outcomes such as time saved, money saved, reduced effort, and reduced fare disputes. Further studies may also explore how the system performs over time and how feedback data can be used to improve fare monitoring and transport transparency, it is recommended to explore adding real time features such as GPS tracking, live fare updates, and traffic conditions to improve the system further. Additionally, future studies can focus on improving the accuracy of fare estimates and making the application more user friendly for different types of users.

Acknowledgment

The researchers gratefully acknowledge the individuals who kindly dedicated their time and provided meaningful insights during the data collection. They also extend their sincere appreciation to the professor who provided guidance throughout the research process.

References

[1] "Metro Manila/Jeepney and UV Express routes," OpenStreetMap Wiki, Dec. 2024., https://wiki.openstreetmap.org/wiki/Metro_Manila/Jeepney_and_UV_Express_routes

[2] Manila LGU releases new fare matrix for tri-wheels," Manila Bulletin, Nov. 7, 2023., https://mb.com.ph/2023/11/7/manila-lgu-releases-new-fare-matrix-for-tri-wheels

[3] A. R. Santos, "Challenges of urban transport development in Metro Manila," in Proc. Eastern Asia Soc. Transp. Studies, vol. 10, 2013., https://easts.info/on-line/proceedings/vol10/pdf/1296.pdf

[4] M. A. Regidor et al., "Travel behavior of Metro Manila public transport commuters," Asian Transp. Studies, vol. 4, no. 2, pp. 460-477, 2016., https://www.jstage.jst.go.jp/article/eastsats/4/2/4_460/_pdf

[5] J. V. F. Cortes et al., "Reforms in Metro Manila's bus transport system hastened by COVID-19," Frontiers Transp. Sustain., 2023., https://pmc.ncbi.nlm.nih.gov/articles/PMC10277182/

[6] Center for People's Development Governance (CPDG), "Commuting in Metro Manila: A daily struggle rooted in inequality," CPDG SDG Watch, Jun. 4, 2025., https://cpdg.ph/commuting-in-metro-manila-a-daily-struggle-rooted-in-inequality/

[7] N. A. Bautista, V. Palad, and J. M. Revilla, "Trike troubles: Manila's new fare matrix frustrates drivers, confuses passengers," The Flame, Apr. 19, 2024., https://abtheflame.net/issues/2024/04/trike-troubles-manilas-new-fare-matrix-frustrates-drivers-confuses-passengers/

[8] PUVMP Funding," LTO Portal, Jan. 2023., https://ltoportal.ph/puvmp-public-utility-vehicle-modernization-program/

[9] LTFRB Approves Fare Increase for Public Transport Starting March 19, 2026," PhilNews, Mar. 17, 2026., https://www.sunstar.com.ph/manila/ltfrb-approves-fare-increase-on-all-puis

[10] Proposed LRT-1 fare hike, unjust amid socioeconomic decline," FEU Advocate, Jul. 21, 2025., https://feuadvocate.net/proposed-lrt-1-fare-hike-unjust-amid-socioeconomic-decline-commuters/

[11] Manila LGU to regulate fare rates for tri-wheels," *Manila Bulletin*, Jul. 7, 2023. https://mb.com.ph/2023/7/7/manila-lgu-to-regulate-fares-for-tri-wheels

[12] Research Study on Tricycle Fare Prices," Univ. Perpetual Help, Aug. 2025., https://www.studocu.com/ph/document/university-of-perpetual-help-system-laguna/health-sciencetechnology-engineering-and-mathematics/research-study-on-tricycle-fare-prices-in-the-philippines-factors-impacts/136705390

[13] "Rolling through Taft: College students' daily ride," *Daily Tribune*, Apr. 20, 2025., https://tribune.net.ph/2025/04/20/rolling-through-taft-college-students-daily-ride

[14] Sakay.ph, "Metro Manila commute directions," 2026., https://www.sakay.ph/

[15] WeSolve Foundation (WSF), "Assessing PUV Modernization Program Industry Consolidation and Financing toward a Just Transition for Transport Workers," *PIDS Policy Paper WSF PP 2024-02*Nov. 10, 2025., https://serp-p.pids.gov.ph/publication/public/view?slug=assessing-puv-modernization-program-industry-consolidation-and-financing-towards-just-transition-for

[16] Fillone, E. M., Tamon, H. L. U., Torrendon, E. J. T., & Osalla, R. S. (2024). The usability of ByaHero: A GPS-based app for navigating public transportation routes in Iloilo City, Philippines. *International Journal of Research in Engineering and Science, 12*(11), 97–100., https://www.ijres.org/papers/Volume-12/Issue-11/121197100.pdf

Authors



Juan Miguel Raphael G. Gaspar is a third-year student in De La Salle-College of Satin Benilde who is currently taking Bachelor of Science in Information Systems. He graduated from the same college under the Arts and Design (ADS) strand. He is taking software engineering as his specialization track and has a strong interest in data analytics and machine learning.



Christian Gil T. Delos Santos is a third-year Bachelor of Science in Information Systems student at De La Salle-College of Saint Benilde. He graduated from St. Andrew's School Inc. Under the Accountancy, Business, and Management (ABM) strand in senior high school. During his free time, he also pursues additional learning through various courses on Udemy to further expand his skills and knowledge.



Malcolm Pyro C. Lopez is a BSIS student in his third year at De La Salle-College of Saint Benilde. He graduated from University of Perpetual Help System DALTA – Molino Campus under the STEM (Science, Technology, Engineering, and Mathematics) strand. He is currently taking data science as his specialization track with strong interests in data analytics and software testing.



Carlo Jezer L. Garcia is a BSIS student in his second year at De La Salle-College of Saint Benilde. He has graduated from Marist School Marikina under the STEM strand. Corporate

Resource Officer at DigiLEAF Inc., training clients in software quality, business analysis, and test automation.M